Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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[C]**June 10-14, 2024** [R]**Ann Arbor, Michigan** [L]**IWDP 2024** [R]**1** [L]**Correspondence to: author@institution.edu**
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authblk [1]First Last of Author A [2]First Last of Author B [1]First Last of Author C [1]Department of Frozen Land, University of South Pole [2]Department of Icicle Engineering, University of North Pole

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The abstract must be submitted by using this **Latex template** or the corresponding **MS Word template** that can be found the meeting website. Submission should be between 300 and 500 words in length. Final submission should be converted and uploaded as a PDF File. LaTeX source files or the original MS Word file will not be accepted. The abstract should be submitted following the file name convention: *last_first.pdf*. Templates margins and text area are set so printed abstracts fit best to the A4 ISO paper size and should not be changed. Font size and type should not be changed. Please also do not change the position of the first line of the title. Abstract submission information can be found on the meeting webpage at **aero.engin.umich.edu/iwdp2024**. Abstract should be submitted for consideration of either the oral presentation or poster; please indicate your selection in the abstract submission form.
Note that headers and footers for the title page are different from those of pages 2 onward. Headers of the title page and footers of other pages must not be changed. Only the left footer of the title page and the headers of other pages must be changed according to the following rules:

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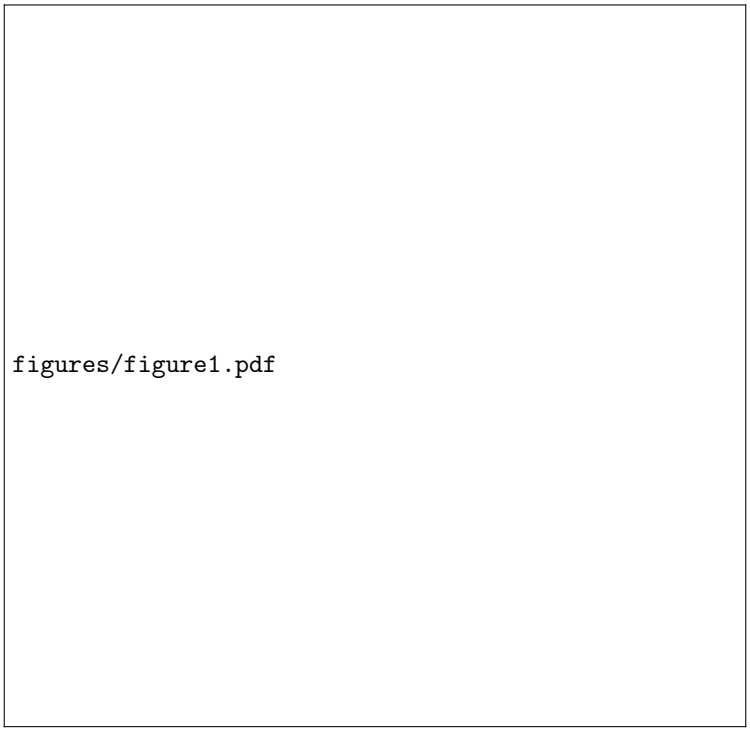
3 Some examples of equations, figures and tables

3.1 An example of an equation

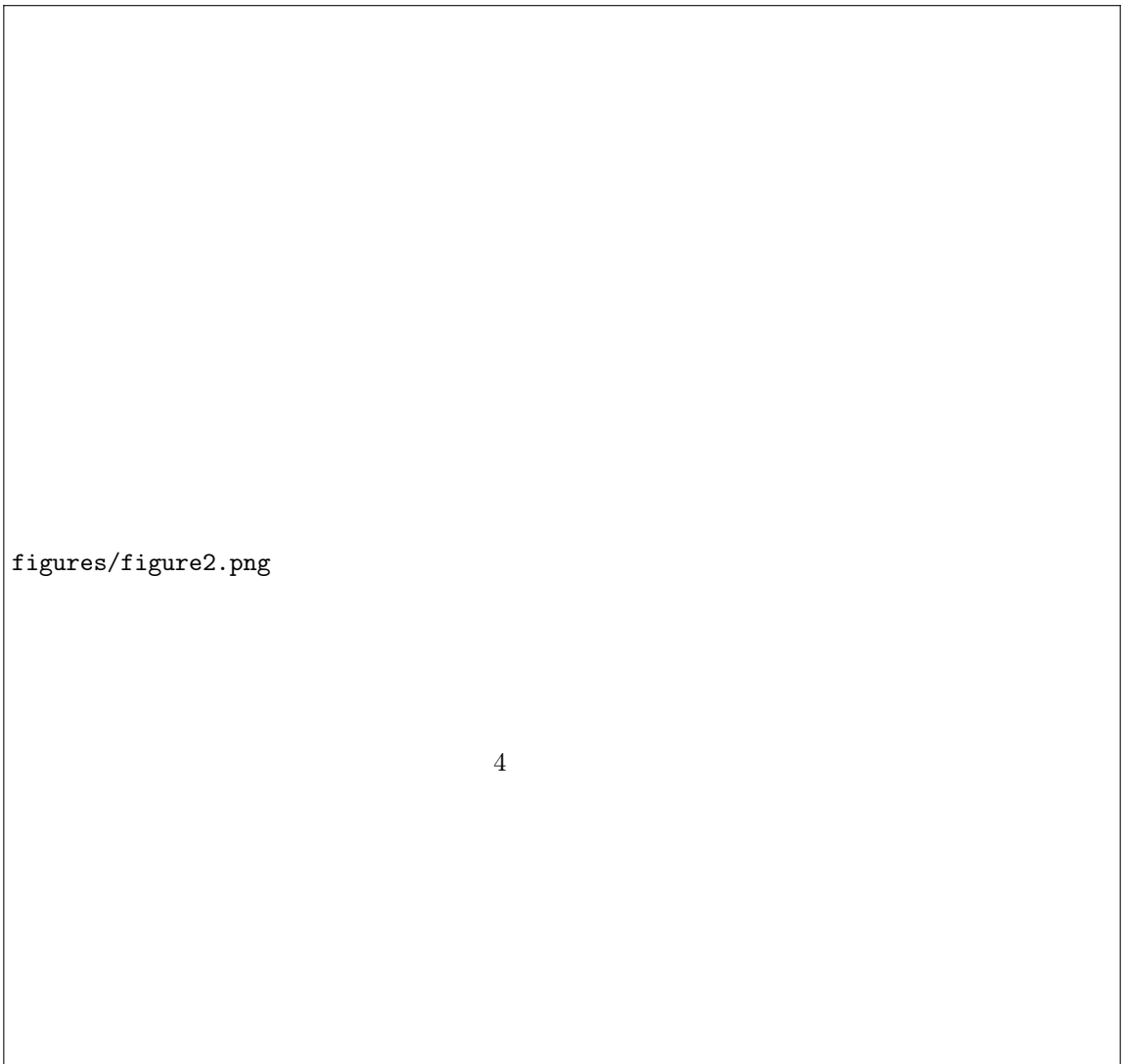
The differential form of conservation equations of mass can be written as: $\mathrm{D}\rho\overline{\mathrm{D}t+\rho\nabla\cdot\mathbf{u}=0}$

3.2 Examples of Figures

Figure 1 and Fig. 3.2 are example of a full width and a test wrapped figure, respectively.
[1-2]



Gambar 1: Example of a full width figure.



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Tabel 1: Description of state location notation in air-breathing engines.

Example of figure with wrapped text.
[3-5]

3.3 Examples of Tables

An example of a table is given in table 1.

4 Conclusions

Please add a conclusion to summarize the main outcomes of this work.